

Real Security Camera — Camera Liberation Feature

Product Specification & Technical Architecture

Version 1.0 | Confidential

Executive Summary

Camera Liberation is a flagship feature that automatically identifies, unlocks, and integrates cloud-locked security cameras into the Real Security Camera platform. No other consumer or enterprise security platform offers this capability.

Value Proposition:

- Homeowners inherit locked cameras when buying a property
 - Businesses replace security providers but keep existing hardware
 - Users refuse to pay ongoing cloud subscriptions for hardware they own
 - This feature solves ALL of these scenarios automatically
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Tier Structure & Monetization

Free Tier

- RTSP cameras (already open) — unlimited
- USB/Webcam — 1 device
- Phone cameras — 1 device
- Local storage only
- 14-day clip retention

Pro Tier — \$14.99/month

- Everything in Free
- Up to 5 cameras
- Camera Liberation — up to 3 liberations/month
- Cloud storage — 60-day retention
- 2 admins
- Priority support

Business Tier — \$39.99/month

- Up to 20 cameras
- Unlimited liberations
- Cloud storage — 180-day retention
- 3 admins + \$4.99/additional admin
- Multi-site support
- API access

Enterprise Tier — \$99.99/month

- Unlimited cameras
 - Unlimited liberations
 - 1-year retention
 - White label option
 - SLA guarantee
 - Dedicated support
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Camera Liberation — Technical Architecture

Phase 1: Detection & Identification

Discovery Agent scans network

- Finds camera by MAC address
- Looks up MAC OUI database
- Identifies manufacturer + model
- Checks if camera is "open" (RTSP works) or "locked"

Locked camera types:

- Alarm.com ADC series (Sercomm/Ambarella hardware)
- Zmodo/Meshare (Sercomm hardware)
- Skybell (proprietary)
- Ring (Amazon cloud)
- Nest (Google cloud)
- Arlo (Netgear cloud)
- Wyze (proprietary)
- Eufy (Anker cloud)

Phase 2: Liberation Methods (in order of attempt)

Method A — Credential Extraction (no flash needed)

Tool: ipctool (OpenIPC project)
How: Connects to camera over network
 Reads running firmware memory
 Extracts RTSP credentials
 Returns username/password
Risk: None — read only
Time: 30 seconds
Success rate: ~70% on Sercomm/Ambarella cameras

Method B — Cloud OAuth Claim

Supported: Alarm.com, Skybell, Ring, Nest, Arlo, Wyze
How: User authenticates with their cloud account
 App claims camera via official API
 Gets streaming token/RTSP credentials
 Stores credentials locally
Risk: None — uses official APIs
Time: 2-3 minutes
Success rate: 95% if user has/creates account

Method C — Firmware Liberation (OpenIPC flash)

Tool: OpenIPC + Coupler
How: Identifies exact SoC chipset via ipctool
 Downloads matching OpenIPC firmware build
 Flashes via camera's own update mechanism
 Camera reboots with open firmware
 RTSP enabled automatically
Risk: Small brick risk (~5%) — disclaimer required
Time: 5-10 minutes
Supported chipsets:

- HiSilicon Hi35xx (most common)
- Ambarella S series
- SigmaStar SSC33x
- Ingenic T20/T31
- XiongMai XM5xx

Success rate: ~85% on supported hardware

Method D — Hardware Bridge (last resort)

How: Install our mobile app on old Android phone
Position phone to view locked camera's screen
Use phone camera as proxy feed
AI upscales and stabilizes the feed

Risk: None

Time: 5 minutes

Success rate: 100% (works on ANY camera)

Note: "If you can see it, we can stream it"

Liberation Wizard — UX Flow

Step 1: Camera Detected

 Locked Camera Detected

ADC-V724 (Alarm.com)

IP: 10.0.0.6

MAC: B8:3A:9D:8A:94:46

Hardware: Sercomm/Ambarella

This camera is cloud-locked.
Real Security Camera can
liberate it for you.

[ Liberate This Camera]

[Skip for now]

Step 2: Disclaimer

 Camera Liberation Disclaimer

- You must own this camera
- Liberation may void warranty
- Firmware flash has small risk
- If bricked: camera needs

replacement (you needed to replace it anyway!)

- Your data stays on YOUR network

By proceeding you confirm you own this hardware.

[ I Own This Camera – Proceed]

[Cancel]

Step 3: Auto-Liberation Attempt

 Liberating Camera...

 Camera identified
ADC-V724 / Sercomm NC450
Ambarella S3L chipset

 Method A: Credential extract...
Connecting via ipctool...
Reading firmware memory...

[] 45%

Step 4: Success

 Camera Liberated!

ADC-V724 (10.0.0.6)

-  RTSP stream active
-  Added to your cameras
-  Recording enabled

rtsp://admin:****@10.0.0.6:554

[▶ Watch Live Now]

Technical Implementation Plan

Agent Updates (stream-agent / discovery-agent)

javascript

```
// liberation-agent.js - new agent
class CameraLiberator {

  async identify(ip, mac) {
    // Look up MAC OUI → manufacturer
    // Run ipctool to get chipset
    // Return { manufacturer, model, chipset, method }
  }

  async liberateMethodA(ip) {
    // ipctool credential extraction
    // Returns { username, password, rtspUrl }
  }

  async liberateMethodB(ip, provider, token) {
    // OAuth claim via provider API
    // alarm.com, ring, nest, etc
    // Returns { rtspUrl, streamToken }
  }

  async liberateMethodC(ip, chipset) {
    // Download OpenIPC firmware for chipset
    // Flash via Coupler
    // Returns { success, rtspUrl }
  }

  async liberateMethodD(ip) {
    // Guide user to use phone as proxy
    // Returns { proxyDeviceId }
  }
}
```

Backend API Endpoints

POST /api/liberation/identify	– identify camera type
POST /api/liberation/start	– start liberation process
GET /api/liberation/status/:id	– get liberation status

```
POST /api/liberation/credential  - save extracted credentials
GET  /api/liberation/supported   - list supported camera models
```

Database Schema Addition

sql

```
CREATE TABLE liberation_attempts (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  organization_id   UUID REFERENCES organizations(id),
  device_id        UUID REFERENCES devices(id),
  camera_ip        VARCHAR(45),
  mac_address      VARCHAR(17),
  manufacturer     VARCHAR(100),
  model            VARCHAR(100),
  chipset          VARCHAR(100),
  method_attempted VARCHAR(1), -- A, B, C, D
  method_succeeded VARCHAR(1),
  status           VARCHAR(20), -- pending, success, failed, bricked
  rtsp_url         TEXT,
  disclaimer_accepted BOOLEAN DEFAULT false,
  created_at       TIMESTAMPTZ DEFAULT NOW(),
  completed_at     TIMESTAMPTZ
);
```

Supported Camera Database

javascript

```
const SUPPORTED_CAMERAS = {
  // MAC OUI prefix → camera info
  'B8:3A:9D': {
    manufacturer: 'Sercomm (Alarm.com/Zmodo OEM)',
    models: ['ADC-V724', 'ADC-V522IR', 'ZP-IBH15'],
    chipset: 'Ambarella S3L',
    liberationMethods: ['A', 'C'],
    openIpcFirmware: 'ambarella-s3l',
    defaultRtspPath: '/live/main',
    notes: 'Factory reset via WPS button 15s'
  },
  '00:26:B9': {
    manufacturer: 'D-Link',
    liberationMethods: ['A'],
    notes: 'Usually open RTSP after reset'
  },
  'B0:C5:54': {
    manufacturer: 'Ring (Amazon)',
    liberationMethods: ['B'],
    oauthProvider: 'ring',
    notes: 'Requires Ring account OAuth'
  },
  '18:B4:30': {
    manufacturer: 'Nest (Google)',
    liberationMethods: ['B'],
    oauthProvider: 'nest',
    notes: 'Requires Google account OAuth'
  },
  'D0:73:D5': {
    manufacturer: 'Arlo (Netgear)',
    liberationMethods: ['B', 'C'],
    oauthProvider: 'arlo'
  },
  '2C:AA:8E': {
    manufacturer: 'Wyze',
    liberationMethods: ['B', 'C'],
    oauthProvider: 'wyze',
    notes: 'Wyze has local RTSP firmware available'
  },
  'A4:DA:32': {
    manufacturer: 'Eufy (Anker)',
    liberationMethods: ['B'],
    oauthProvider: 'eufy'
  }
}
```

```
}  
};
```

Pricing Strategy for Liberation Feature

Liberation as Upsell

Free tier sees locked camera →

"🔒 This camera is locked. Upgrade to Pro to liberate it with one click."

Pro tier: 3 liberations/month included

Business tier: Unlimited liberations

Enterprise: Unlimited + white label liberation tool

One-Time Liberation Purchases

For free tier users who don't want to upgrade:

Single camera liberation: \$4.99

3-pack liberation: \$9.99

This creates revenue WITHOUT requiring subscription

Liberation as Enterprise Feature

For property management companies, Airbnb hosts, real estate agents who buy/sell properties:

"Property Liberation Pack" – \$49.99 one-time

Liberates ALL cameras found on a single network

Generates report of all cameras found + liberated

Perfect for new property owners

Competitive Moat

No competitor offers this because:

1. It requires deep hardware knowledge
2. It requires maintaining a camera database

3. Legal grey area (we solve with disclaimer)
4. Complex multi-method fallback system
5. Requires local agent (we have one)

This is 12-18 months ahead of any competitor.

Phase Roadmap

Phase 4A (Next 2 weeks) — Foundation

- ☐ ipctool integration in discovery-agent
- ☐ MAC OUI camera database (50 most common models)
- ☐ Liberation API endpoints (backend)
- ☐ Liberation attempts DB table
- ☐ Basic UI in Discover tab

Phase 4B (Month 2) — Methods A & B

- ☐ Method A: ipctool credential extraction
- ☐ Method B: Alarm.com OAuth flow
- ☐ Method B: Ring OAuth flow
- ☐ Method B: Nest OAuth flow
- ☐ Liberation Wizard UI (full flow)
- ☐ Success/failure tracking

Phase 4C (Month 3) — Method C

- ☐ OpenIPC firmware database
- ☐ Coupler integration in agent
- ☐ Automated flash for top 20 chipsets
- ☐ Brick recovery guide
- ☐ Liberation history in dashboard

Phase 4D (Month 4) — Method D & Mobile

- ☐ Method D: Phone proxy bridge
 - ☐ React Native liberation wizard
 - ☐ Push notifications for liberation status
 - ☐ Enterprise property scan feature
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Legal Framework

Disclaimer Language

"Camera Liberation modifies firmware on hardware you own.
By proceeding you confirm:
(1) You own or have explicit permission to modify this device
(2) You understand warranty may be voided
(3) Real Security Camera is not liable for device damage
(4) This tool is for personal security use only

Liberation is protected under:
- Right to Repair principles
- DMCA Section 1201 exemptions for security research
- Personal property rights"

Terms of Service Addition

Users agree not to use Camera Liberation on:
- Devices they do not own
- Commercial/government surveillance without consent
- Any device where modification is illegal in their jurisdiction

Success Metrics

Metric	3 Month Target	6 Month Target
Cameras liberated	500	5,000
Supported models	25	100
Liberation success rate	70%	85%
Upsell conversion	15%	25%
Monthly liberation revenue	\$2,500	\$15,000

